

Ex:14

PCAPXRAY TOOL

AIM

PcapXray is a network forensics tool that visualizes a packet capture (PCAP) offline as an interactive network diagram to speed packet-level investigation, identify devices/hosts, highlight important or suspicious communications (including Tor and potential malicious traffic), and extract files transferred over the capture

ALGORITHM:

- 1.Install PcapXray dependencies using: `sudo pip3 install -r requirements.txt`
- 2.Launch the tool with: `sudo python3 PcapXray.py`
- 3.Open the web interface and upload the given PCAP file.
- 4.Click Analyze to process the packet data.
- 5.View the generated network topology and communication graph.
- 6.Examine hosts, protocols, and suspicious connections.
- 7.Save the generated report and diagrams for documentation

OUTPUT:

```
(kali@kali)-[~]
└─$ sudo git clone https://github.com/Srinivas11789/PcapXray.git
[sudo] password for kali:
Cloning into 'PcapXray' ...
remote: Enumerating objects: 1704, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 1704 (delta 3), reused 9 (delta 2), pack-reused 1689
Receiving objects: 100% (1704/1704), 115.75 MiB | 1.14 MiB/s, done.
Resolving deltas: 100% (975/975), done.
```

```
(kali@kali)-[~]
└─$ sudo apt install graphviz
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
(kali@kali)-[~]
└─$ sudo apt install python3-pil python3-pil.imagetk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libev4 libfmt8 libhttp-server-simple-perl libilmbase25 liblerc3 libopenexr25
  libpoppler118 libpython3.9-minimal libpython3.9-stdlib libsvtav1enc0 libwebsockets16
  linux-image-5.18.0-kali5-amd64 python3-dataclasses-json python3-limiter
  python3-marshmallow-enum python3-mypy-extensions python3-responses python3-spyse
  python3-token-bucket python3-typing-inspect python3.9 python3.9-minimal
```

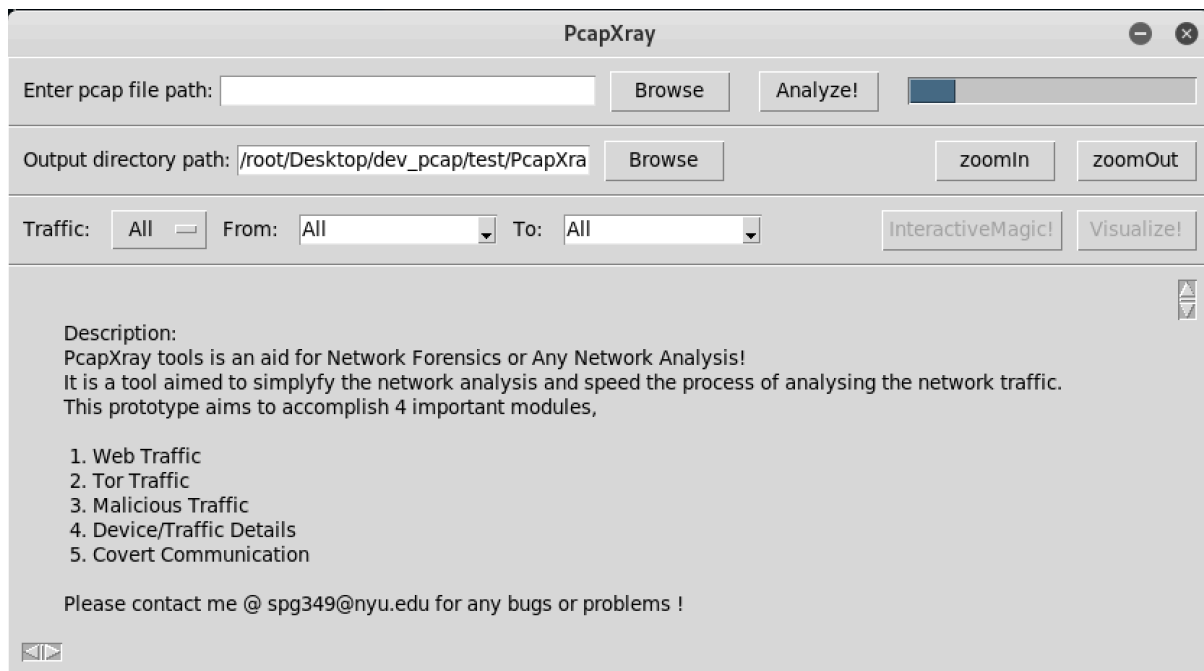
```
(kali@kali)-[~]
└─$ ls
Desktop    Downloads  Music      Pictures   Templates  Videos
Documents  get-pip.py PcapXray   Public     tmp
```

```
(kali@kali)-[~]
└─$ cd PcapXray
```

```
(kali@kali)-[~/PcapXray]
└─$ ls
_config.yml  Dockerfile  logo.png   requirements.txt  Samples  Test
Design       LICENSE    README.md  run.sh           Source
```

```
(kali@kali)-[~/PcapXray]
└─$ pip3 install -r requirements.txt
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: scapy in /usr/lib/python3/dist-packages (from -r requirements.txt (line 4)) (2.4.5+g9420c22)
Collecting pyshark
  Downloading pyshark-0.5.3-py3-none-any.whl (41 kB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 41.1/41.1 kB 398.6 kB/s eta 0:00:00
Requirement already satisfied: ipwhois in /usr/lib/python3/dist-packages (from -r requirements.txt (line 12)) (0.15.1)
```

```
(kali@kali)-[~/PcapXray]
└─$ python3 Source/main.py
```



RESULT:

Network forensics succeeded with PcapXray: hosts, protocols, and communication flows were visualized from the submitted PCAP.